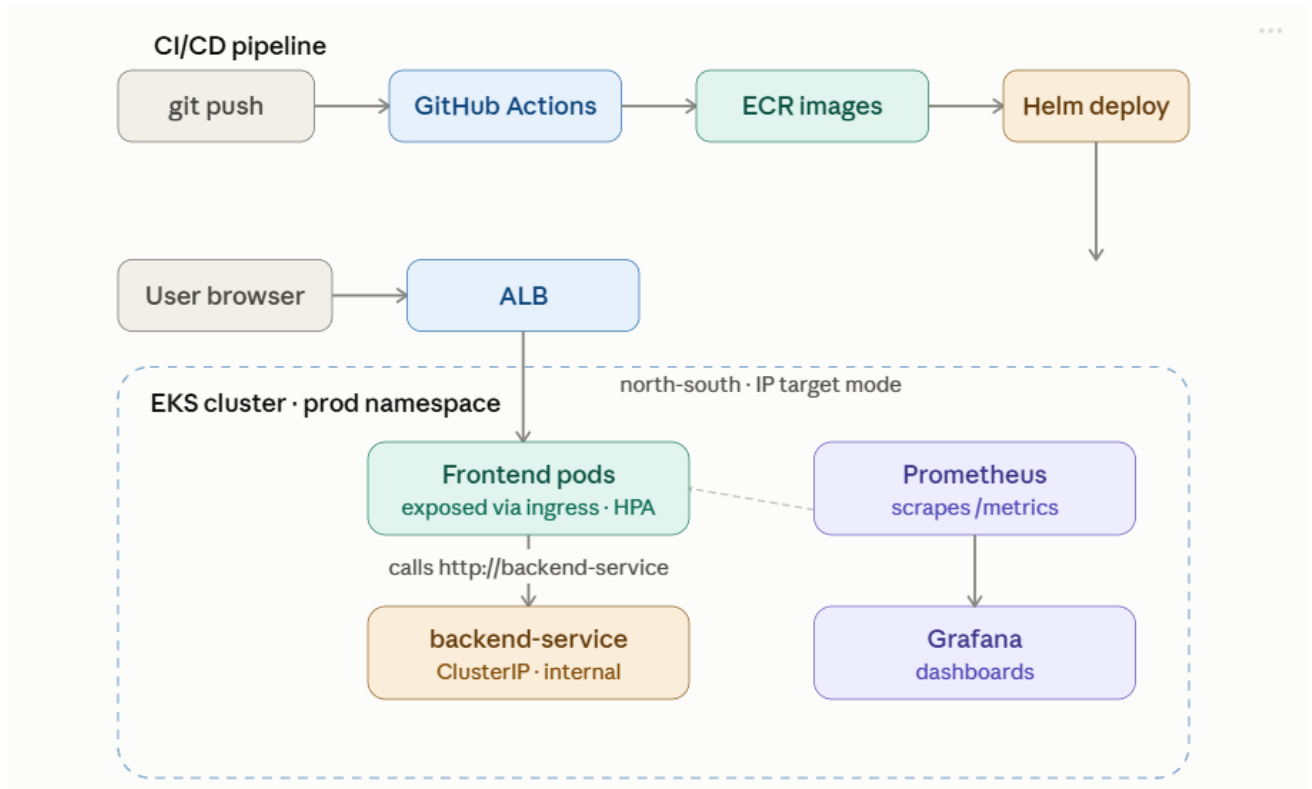


EKS CI/CD Demo Project - A two-tier microservices application on AWS EKS, deployed automatically by a CI/CD pipeline, exposed through an ALB, with full observability via Prometheus and Grafana.



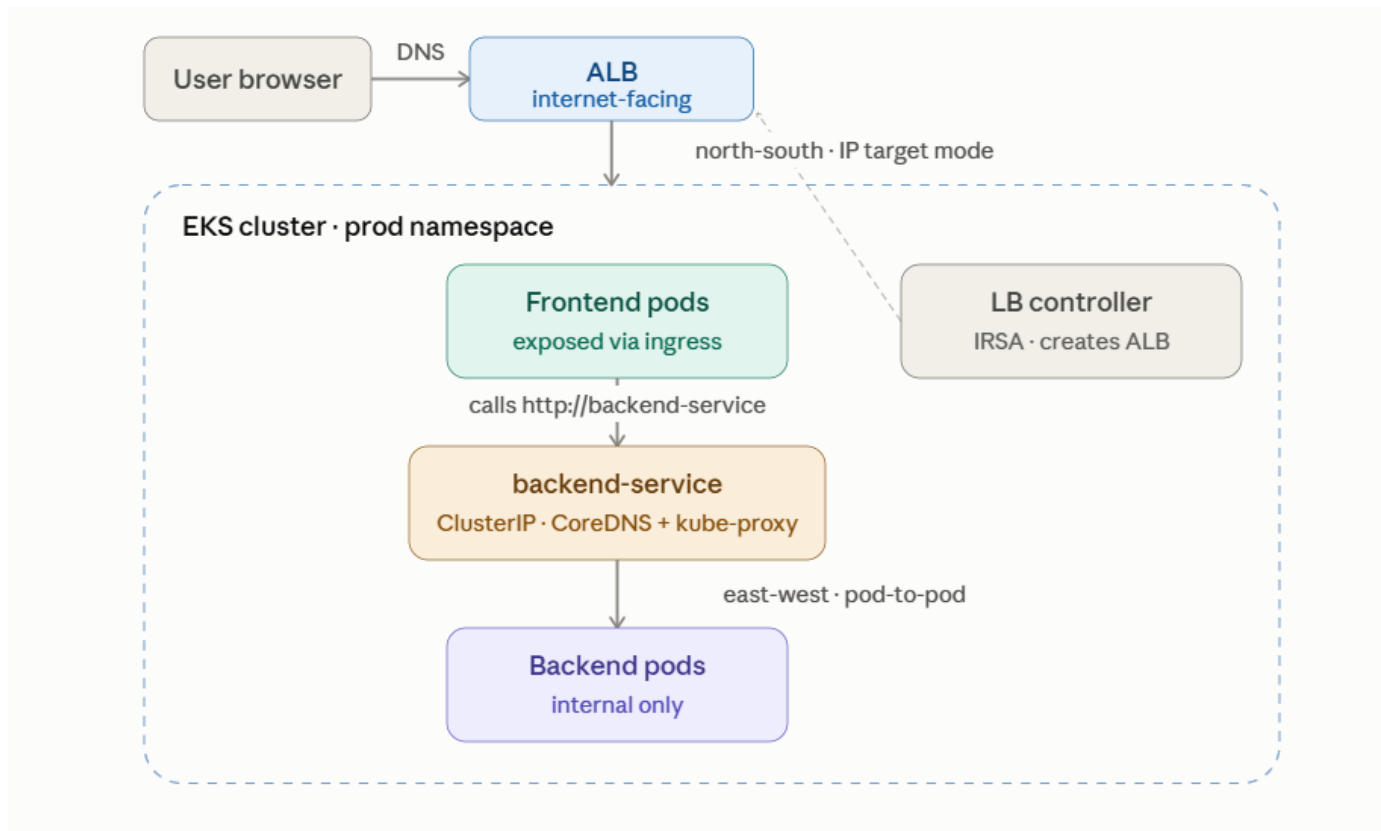
What I have implemented in this project:

- The running app(using Ingress)
- Self-healing
- Autoscaling
- CI/CD pipeline(github actions deployed through Helm)
- Monitoring & Alerting(Prometheus and grafana)

Prometheus custom metrics(exposed through /metrics):

- http_requests_total - Total HTTP requests received on frontend pods
- backend_calls_total - Total calls made to the backend
- backend_errors_total - Total failed backend calls

Internal pod-to-pod communication:



Browser → DNS → ALB → frontend pod → frontend calls backend → CoreDNS → backend ClusterIP → kube-proxy → backend pod → response.